DEPARTMENT OF COMPUTER SCIENCE / INFORMATION TECHNOLOGY
Faculty of Computing & Applied Sciences
University of Southern Punjab, Multan

Web-Based Centralized Platform That Optimizes Food Waste Redistribution by Linking Restaurants with Registered Charitable Organizations

FINAL YEAR PROJECT (FYP) • BSCS SESSION 2024–2026 • SHAREMEAL ECOSYSTEM

DESCRIPTION

Food wastage is one of the most pressing socio-economic challenges in Pakistan, where millions endure severe nutritional deprivation while banquet halls, hotels, and restaurants discard metric tons of edible surplus food daily. The **ShareMeal Platform** is an enterprise web-based centralized solution engineered using **ASP.NET Core 8.0, C# 12, Entity Framework Core, and Google Gemini AI**. The system provides real-time digital linking between commercial surplus food donors and vetted charitable welfare NGOs. By digitizing food inventory listings, automating pickup dispatching, enforcing strict food safety time-windows, and providing AI-powered operational assistance, ShareMeal bridges the critical logistics gap to eradicate food waste and feed impoverished communities sustainably.

SCOPE OF THE SYSTEM

- **Nationwide Cloud Accessibility:** Responsive web platform functioning seamlessly across desktop, tablet, and mobile devices without requiring app store installation.
- **Multi-Role Role-Based Architecture:** Segregated, secure portals for Food Donors (Restaurants), Claiming Charities (NGOs), and System Administrators.
- **Real-Time Food Listing & Matching:** Dynamic surplus inventory listing containing meal portions, preparation timestamp, preservation state, and expiry countdown.
- **Atomic Concurrency Claim Engine:** Database-level transaction locking preventing duplicate claims or race conditions between competing charities.
- **Bilingual Multimodal Gemini AI (MealBot):** 24/7 intelligent voice-enabled chatbot providing automated SOP instructions, platform navigation, and meal safety assistance.
- **Interactive Geolocation Mapping:** Integration with Leaflet.js and GPS routing for pinpointing donor addresses, transit estimation, and distribution verification.

FUNCTIONAL REQUIREMENTS

- User registration, login, and profile management with hashed credentials (ASP.NET Core Identity).
- Restaurant / Donor Module: Add, edit, manage, and monitor surplus food listings with expiry deadlines.
- Charity / NGO Module: Real-time marketplace feed, location filtering, instant donation claim reservation.
- Strict Verification Pipeline: Admin approval workflows verifying government NGO credentials and NTN.
- Status Lifecycle Engine: Automated transitions across *Available* → *Claimed* → *In-Transit* → *Delivered*.
- Food Safety & Hygiene SOPs: Mandatory pre-claim verification of preparation time and food temperature.
- MealBot AI Integration: Natural language processing in Roman Urdu, Urdu, and English with Web Speech API.

COMPARATIVE ANALYSIS

Operational Feature	Manual Calling	WhatsApp Groups	ShareMeal Platform
Real-Time Matching	No	Partial	Yes (Instant)
Double-Claim Prevention	No	No (High Chaos)	Yes (Atomic Lock)
Verified NGO Authorization	No	No	Yes (Admin Vetted)
Food Safety & Expiry Tracking	No	No	Yes (Auto Countdown)
24/7 AI-Powered Assistance	No	No	Yes (Gemini AI)
Voice Accessibility (Bilingual)	No	No	Yes (Web Speech)
Live GPS Route Mapping	No	Manual Links	Yes (Leaflet.js)
Audit Trail & Impact Reports	No	No	Yes (Automated)

RELATIONAL SCHEMA DIAGRAM



Figure: ShareMeal Normalized Relational Entity Relationship Diagram (ERD)

TOOLS & TECHNOLOGIES

Hardware Tools

- **Processor:** Intel Core i7 / i5 Multi-Core
- **Memory:** 16GB DDR4 High-Speed RAM
- **Storage:** 512GB NVMe SSD
- **Display:** 1080p / 4K UHD Workstation
- **Testbed:** iOS Safari & Android Chrome

Software Tools

- **Backend:** ASP.NET Core 8.0 & C# 12
- **ORM & DB:** Entity Framework Core 8
- **Frontend:** Modern HTML5, TailwindCSS
- **AI Engine:** Google Gemini Flash API
- **Maps & Audio:** Leaflet & Web Speech

- System Analytics & Reporting: Comprehensive PDF and visual telemetry for meals saved and carbon offset.

- **Cloud / DevOps:** Docker & Railway PaaS

METHODOLOGY (AGILE RAD MODEL)

The project followed the **Rapid Application Development (RAD)** methodology under an Agile iterative framework, emphasizing rapid prototyping, user feedback loops, and automated cloud deployments:



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